Kevin Ho

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Education

Mississippi State University, Starkville, MS *August 2023 – May 2026*

Bachelor of Science, Computer Science *GPA: 4.0*
Bachelor of Science, Mathematics *GPA: 4.0*
Relevant Courses: Data Structures, Algorithms, Artificial Intelligence, PDEs, Numerical Analysis, Linear Algebra, Real Analysis, Probability and Statistics

Experience

Medical Diagnosis Software Development, Starkville, MS *July 2024 – Present*
Undergraduate Research Assistant

- Developed AI-driven brain tumor detection software using **Convolutional Neural Networks (CNNs)** on MRI data, achieving over **95%** precision to assist early diagnosis.
- Collaborated with medical professionals to integrate the software into clinical workflows at **UMMC**, enhancing efficiency in radiology imaging analysis.

Dynamics Across Multiple Networks, Starkville, MS *September 2024 – Present*
Undergraduate Research Assistant

- Analyzed vulnerabilities and behavior in complex networks like **rumor propagation** and **disease transmission** using **graph theory** and Python-based simulations.
- Contributed to developing a unified framework for improving **system resilience** and **risk mitigation**, with findings presented at research symposiums.

Projects

Hand Tracking Using Computer Vision *OpenCV, Numpy, Mediapipe, Sklearn*

- Developed a real-time hand tracking system using **OpenCV** and **Mediapipe** to detect and track hand movements.
- Used **Sklearn** to fine-tune classifier models for gesture categorization, and optimized frame processing speed to ensure fluid real-time interactions without lag.
- Implemented a user-friendly way for users to train the model to add their own gesture.

DiAlignment: LLM Output Modification with Activation Vector Scaling *Python, Flask, Gemini API*

- Implemented dynamic scaling of topic weights in model outputs, providing a novel alternative to traditional prompt engineering methods.
- Evaluated model vulnerabilities to **rejection white-box attacks**, establishing benchmarks for AI alignment assessments.
- Tested model susceptibility to rejection white-box attacks, providing a benchmark for AI Alignment
- ***Awarded Best Use of Streamlit at the AI ATL Hackathon for innovative application design.***

University Building Image Classification Model *Python, TensorFlow*

- Developed an image classification model using TensorFlow and a custom ResNet-based deep learning architecture to classify university buildings, achieving **98.71%** accuracy on a dataset of 10,000+ images.
- Implemented custom callbacks for checkpointing and validation tracking in Google Colab, fine-tuned learning rates, and utilized advanced data augmentation techniques to prevent overfitting and improve model generalization.
- ***Awarded 1st place for Mississippi State Campus Vision Competition***

Extracurriculars

Chess Club, Mississippi State University *President, August 2023 – Present*

- Organized weekly meetings, chess events, and engaged the campus community in competitive and recreational play.
- Coordinated tournaments with local schools and universities, managing event logistics and securing sponsorships, raising funds to support club activities and tournaments.

ACM Competitive Programming Club, Mississippi State University *Vice President, August 2023 – Present*

- **2024 ICPC Southeast Regionals Gold Medalist**
- Organized coding competitions and practice sessions, helping prepare members for national programming contests like **ICPC** by creating a structured training curriculum.
- Mentored students in algorithm design and data structures, emphasizing practical problem-solving strategies.

Skills

Programming Languages: Python, C/C++, Java, Assembly
Machine Learning: TensorFlow, PyTorch, NumPy, Hugging Face, Scikit-learn
Database Technologies: SQL, SQLite, MongoDB
Web Development: React, Google Cloud, Flask, Streamlit, Cloudflare
Software & Tools: Git, Jetbrains, Matplotlib, Jupyter, Visual Studio Code